

Participantes:

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Repo:

<https://github.com/dsrdantas/TC4-TM>

Video: <https://github.com/dsrdantas/TC4-TM/blob/main/tc4-video.mp4>

Justificativa técnica:

1. Os serviços geram dados de observabilidade

Cada serviço (auth, flag, targeting, evaluation, analytics) é instrumentado para registrar automaticamente o que está acontecendo: qual endpoint foi chamado, quanto tempo demorou, se houve erro, qual query foi feita no banco. Esses registros são chamados de spans (para traces), métricas e logs.

2. O OTel Collector recebe tudo num ponto central

Em vez de cada serviço enviar dados diretamente para o New Relic, todos enviam para o OTel Collector — que funciona como um roteador dentro do cluster. Ele recebe os dados, adiciona informações de contexto (nome do cluster, ambiente) e decide para onde mandar cada tipo:

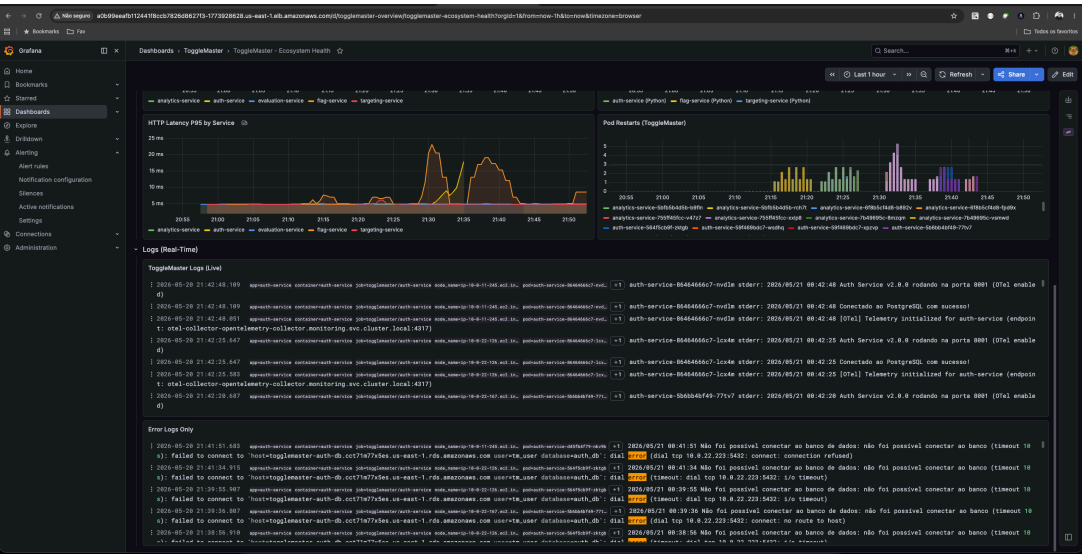
3. O New Relic recebe e monta os traces

O Collector envia os spans para o New Relic via HTTPS. Como cada span carrega o ID do trace pai, o New Relic consegue montar a árvore completa de uma requisição — mostrando que o /evaluate chamou o flag-service, que consultou o PostgreSQL, tudo dentro do mesmo trace.

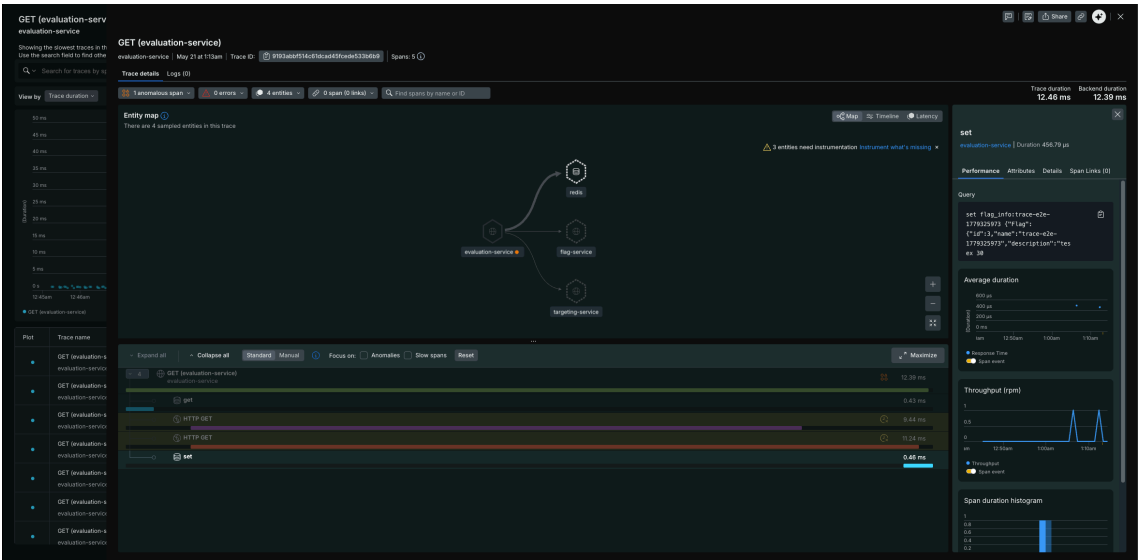
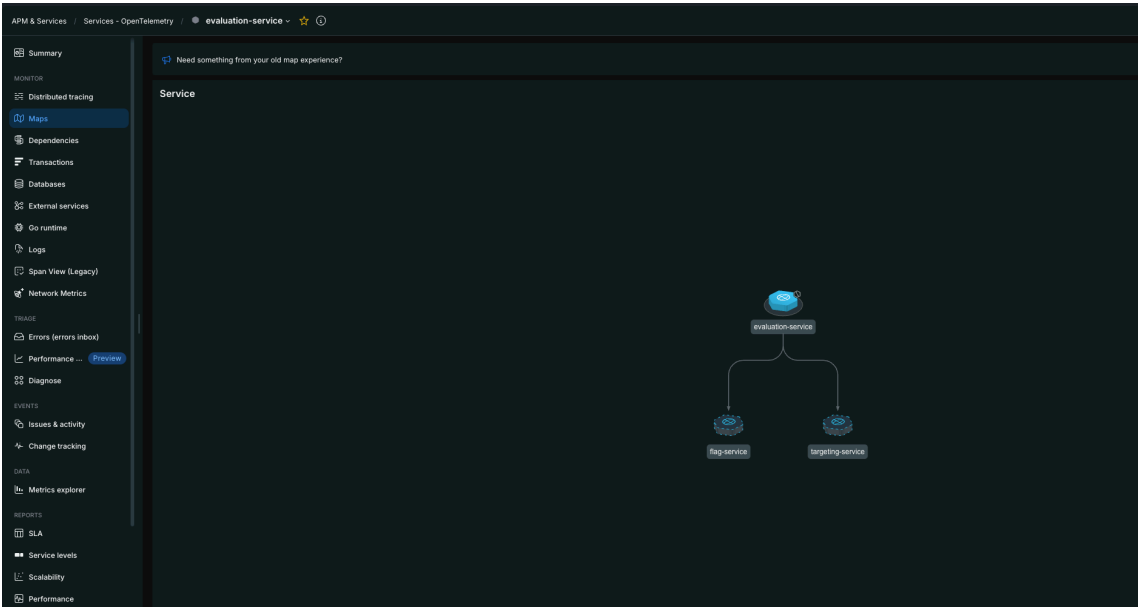
O benefício do Collector no meio é que os serviços não precisam saber nada sobre o New Relic — eles só falam com o Collector. Se amanhã quiser trocar o destino, muda só a configuração do Collector, sem tocar no código dos serviços.

Evidencias

Grafana



Trace Newrelic



Self-Healing

github.com/isrdantas/TC4-1M/actions/runs/2619846850/job/77083004233

Warning

Auto-Remediate Alert

succeeded 7 minutes ago in 51s

Set up job

Checkout

Determine target service

Configure AWS credentials

Configure kubectl

Check current pod status (before)

Execute rollout restart

Check pod status (after)

Notify Discord

1

Run echo >>> Executing: kubectl rollout restart deployment/auth-service -n togglemaster

16

>>> Executing: kubectl rollout restart deployment/auth-service -n togglemaster

17

deployment.apps/auth-service restarted

18

>>> Waiting for rollout to complete...

19

Waiting for deployment "auth-service" rollout to finish: 1 out of 2 new replicas have been updated...

20

Waiting for deployment "auth-service" rollout to finish: 1 out of 2 new replicas have been updated...

21

Waiting for deployment "auth-service" rollout to finish: 1 out of 2 new replicas have been updated...

22

Waiting for deployment "auth-service" rollout to finish: 1 out of 2 new replicas have been updated...

23

Waiting for deployment "auth-service" rollout to finish: 1 old replicas are pending termination...

24

Waiting for deployment "auth-service" rollout to finish: 1 old replicas are pending termination...

25

deployment "auth-service" successfully rolled out

1

Run echo >>> Pod status (AFTER restart) >>>

16

>>> Pod status (AFTER restart) >>>

17

NAME READY STATUS RESTARTS AGE IP NODE NOMINATED NODE READINESS GATES

18

auth-service-86d6d6c7-1cx4e 1/1 Running 0 46s 18.0.22.140 ip-10-0-22-126.ec2.internal <none> <none>

19

auth-service-86d6d6c7-media 1/1 Running 0 23s 18.0.11.197 ip-10-0-11-240.ec2.internal <none> <none>

20

21

>>> Deployment status >>>

22

NAME READY UP-TO-DATE AVAILABLE AGE

23

auth-service 2/2 2 2 77m

1

Run if ["success" = "success"]; then

48

Discord notification sent.

Summary

Post Configure AWS credentials

Post Checkout

Complete job

minutes. **Alert:** PodNotReady - WARNING **Summary:** Pod targeting-service-ddc56d69f-fmqng is not ready **Description:** Pod targeting-service-ddc56d69f-fmqng in namespace togglemaster has been not ready for 5 minutes.

CRITICAL: PodCrashLooping
Alert: PodCrashLooping - CRITICAL **Summary:** Pod auth-service-5b6bb4bf49-77tv7 is crash-looping **Description:** Pod auth-service-5b6bb4bf49-77tv7 (container: auth-service) in namespace togglemaster has been in CrashLoopBackOff for more than 2 minutes. **Alert:** PodCrashLooping - CRITICAL **Summary:** Pod auth-service-d45fb4f79-nkv9k is crash-looping **Description:** Pod auth-service-d45fb4f79-nkv9k (container: auth-service) in namespace togglemaster has been in CrashLoopBackOff for more than 2 minutes. **Alert:** PodCrashLooping - CRITICAL **Summary:** Pod auth-service-d45fb4f79-wttl2 is crash-looping **Description:** Pod auth-service-d45fb4f79-wttl2 (container: auth-service) in namespace togglemaster has been in CrashLoopBackOff for more than 2 minutes.
quarta-feira, 20 de maio de 2026 às 21:43

alerts APP 21:43

Self-Healing SUCCESS

Automated remediation for alert PodCrashLooping

Service	Action	Namespace
auth-service	rollout restart	togglemaster

Triggered By

PodCrashLooping

Timestamp

2026-05-21T00:42:18Z

Workflow Run

[View Logs](#)

ToggleMaster Self-Healing | Phase 4

Alertmanager APP 21:46

WARNING: PodNotReady

Alert: PodNotReady - WARNING **Summary:** Pod auth-service-5b6bb4bf49-77tv7 is not ready **Description:** Pod auth-service-5b6bb4bf49-77tv7 in namespace togglemaster has been not ready for 5 minutes. **Alert:** PodNotReady - WARNING **Summary:** Pod auth-service-d45fb4f79-nkv9k is not ready **Description:** Pod auth-service-d45fb4f79-nkv9k in namespace togglemaster has been not ready for 5 minutes. **Alert:** PodNotReady - WARNING **Summary:** Pod auth-service-d45fb4f79-wttl2 is not ready **Description:** Pod auth-service-d45fb4f79-wttl2 in namespace togglemaster has been not ready for 5 minutes.



alerts APP 21:43

**Self-Healing SUCCESS**

Automated remediation for alert **PodCrashLooping**

Service	Action	Namespace
auth-service	rollout restart	togglemaster

Triggered By
PodCrashLooping

Timestamp
2026-05-21T00:42:18Z

Workflow Run
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ToggleMaster Self-Healing | Phase 4



Alertmanager APP 21:46

WARNING: PodNotReady

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WARNING: KubePodCrashLooping

Alert: KubePodCrashLooping - WARNING *Summary:* Pod is crash looping. *Description:* Pod togglemaster/auth-service-d45fb4f79-nkv9k (auth-service) is in waiting state (reason: "CrashLoopBackOff") on cluster .

CRITICAL: PodCrashLooping

Alert: PodCrashLooping - CRITICAL *Summary:* Pod auth-service-564f5cb9f-zktgb is crash-looping *Description:* Pod auth-service-564f5cb9f-zktgb (container: auth-service) in namespace togglemaster has been in CrashLoopBackOff for more than 2 minutes. *Alert:* PodCrashLooping - CRITICAL *Summary:* Pod auth-service-5b6bb4bf49-77tv7 is crash-looping *Description:* Pod auth-service-5b6bb4bf49-77tv7 (container: auth-service) in namespace togglemaster has been in CrashLoopBackOff for more than 2 minutes.

WARNING: KubePodCrashLooping

Alert: KubePodCrashLooping - WARNING *Summary:* Pod is crash looping. *Description:* Pod togglemaster/auth-service-d45fb4f79-nkv9k (auth-service) is in waiting state (reason: "CrashLoopBackOff") on cluster .

Activity over last 7 days

Service	Title	Time	Activity
Default Service	[#5] [PodCrashLooping] Alert fired in togglemaster	at 9:47 PM	Resolved through the integration API
Default Service	[#5] [PodCrashLooping] Alert fired in togglemaster	at 9:37 PM	Triggered through the API Description: [PodCrashLooping] Alert fired in togglemaster (View Message)